Reflection: Importance of Path Planning and Trade-Offs Between Global and Local Planners

Path planning is a fundamental component of robot autonomy because it enables a robot to move from its current location to a desired goal **safely**, **efficiently**, and **without human intervention**. Without effective path planning, a robot cannot navigate dynamic or unknown environments, avoid obstacles, or optimize travel distance and time. It bridges the gap between sensing the environment and acting upon it, making it crucial for applications like delivery robots, warehouse automation, and service robotics.

In practice, path planning involves two complementary components: the **global planner** and the **local planner**. Each has distinct roles and trade-offs.

The **global planner** is responsible for generating a complete path from the start to the goal based on a static map. It's good for creating long-term, optimal paths but assumes the environment is fully known and static. Its main limitation is that it doesn't handle unexpected changes or moving obstacles in real-time.

The **local planner**, on the other hand, adjusts the robot's motion in real-time to react to dynamic obstacles or small deviations. It ensures safe and smooth movement, even if the environment changes. However, it has limited foresight and might get stuck in local minima or fail to consider the entire path context.

Overall, an effective navigation system balances both: **global planning for long-term strategy**, and **local planning for short-term safety and adaptability**. Understanding and tuning both layers—like adjusting inflation parameters—are key to robust and intelligent robot behavior.